

Extracted Features In Model

No.	Quantity	Representation
Molecular structure		
1	Atomic Cartesian positions	Centred Cartesian coordinates and graph geometry
2	Covalent bond table	Covalent edges with order and category labels
3	Residue table and sequence links	Residue labels and sequence relations
4	Atom-to-residue index	Atom-residue index and broadcast map
5	Atom chemical-category table	14 categorical labels and charge-sum scalar
6	Enriched atom role	Learned categorical embedding
7	Enriched hybridisation class	Learned categorical embedding
Geometry and secondary structure		
26	N-C-alpha-C angle	Residue scalar broadcast to nodes
27	N-C-alpha-C validity	Residue validity mask broadcast to nodes
28	N-C-alpha-C-beta angle	Residue scalar broadcast to nodes
29	N-C-alpha-C-beta validity	Residue validity mask broadcast to nodes
30	C-beta-C-alpha-C angle	Residue scalar broadcast to nodes
31	C-beta-C-alpha-C validity	Residue validity mask broadcast to nodes
32	Previous-C-N-C-alpha angle	Residue scalar broadcast to nodes
33	Previous-C-N-C-alpha validity	Residue validity mask broadcast to nodes
34	C-alpha-C-next-N angle	Residue scalar broadcast to nodes
35	C-alpha-C-next-N validity	Residue validity mask broadcast to nodes
36	Backbone and side-chain torsion cosines	Six torsion-cosine node scalars
37	Torsion validity mask	Six binary node applicability flags
38	Backbone and side-chain torsion sines	Six torsion-sine node scalars
39	Peptide omega cosine	Omega-cosine node scalar
40	Peptide omega validity	Binary node validity mask
41	Peptide omega sine	Omega-sine node scalar
42	X-Pro peptide-bond flag	Binary node indicator
43	Atomic pyramidalisation	Node invariant scalar
44	Pyramidalisation validity	Binary node applicability flag
45	DSSP eight-state assignment	Eight one-hot node scalars
46	DSSP hydrogen-bond energies	Four node invariant scalars
47	DSSP observation mask	Binary node validity mask
48	DSSP polyproline-II quantity	Ternary node scalar

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AIMNet2		
8	AIMNet2 per-atom embedding	256 node invariant scalars
MOPAC atomic and electrostatic		
9	Coulson atomic charge	Node invariant scalar
10	Atomic bond valency	Node invariant scalar
11	Atomic s-shell population	Node invariant scalar
12	Atomic p-shell population	Node invariant scalar
13	Atomic d-shell population	Node invariant scalar
14	Lewis-bond partner count	Node invariant scalar
15	Total Coulson-charge electric field	Node Cartesian vector (T1)
16	Total Coulson-charge field gradient	Node rank-2 tensor (T2)
17	Backbone-source Coulson field	Node Cartesian vector (T1)
18	Side-chain-source Coulson field	Node Cartesian vector (T1)
19	Aromatic-source Coulson field	Node Cartesian vector (T1)
20	Backbone-source Coulson field gradient	Node rank-2 tensor (T2)
21	Side-chain-source Coulson field gradient	Node rank-2 tensor (T2)
22	Aromatic-source Coulson field gradient	Node rank-2 tensor (T2)
23	Coulson-field scalar summaries	Four node scalars and one applicability flag
24	Aromatic field projected along H-parent bond	Node scalar and one applicability flag
25	Aromatic source-atom count	Node invariant count scalar
Fixed McConnell responses		
49	Fixed peptide C-N response	Calibrated node T0 scalar
50	Fixed other-backbone response	Calibrated node T0 scalar
51	Fixed other-side-chain response	Calibrated node T0 scalar
52	Fixed backbone X-H response	Calibrated node T0 scalar
53	Fixed side-chain X-H response	Calibrated node T0 scalar
Wiberg-weighted responses		
54	Wiberg-weighted peptide C=O response	Two node T0+T1+T2 blocks
55	Wiberg-weighted other-backbone response	Node T0+T1+T2 block
56	Wiberg-weighted side-chain C=O response	Node T0+T1+T2 block
57	Wiberg-weighted other-side-chain response	Node T0+T1+T2 block
58	Wiberg-weighted disulfide response	Node T0+T1+T2 block
59	Wiberg-weighted aromatic response	Node T0+T1+T2 block
60	Wiberg-weighted backbone X-H response	Node T0+T1+T2 block
61	Wiberg-weighted side-chain X-H response	Node T0+T1+T2 block
62	Wiberg-weighted S-H response	Node T0+T1+T2 block

No.	Quantity	Representation
Near-field MOPAC responses		
63	Accepted and rejected near-field counts	Two node count scalars
64	Nearest peptide-carbonyl distance	Node distance scalar and applicability flag
65	Nearest peptide C-N distance	Node distance scalar and applicability flag
66	Nearest peptide-carbonyl direction	Node Cartesian vector (T1)
67	Nearest peptide-carbonyl response tensor	Node T0+T1+T2 block
68	Nearest peptide C-N response tensor	Node T0+T1+T2 block
MOPAC density and orbitals		
69	Atomic-orbital block width per atom	Integer AO-width reduction support
70	Atomic AO-density matrix	Node T1+T2 reductions
71	Full-precision Wiberg pair table	Pair scalar; five node scalars; node T1+T2
72	AO-density pair index table	Pair-index reduction support
73	Pair AO-density blocks	11 pair scalars; three T1; one T2
74	Lewis-bond partner indices	Binary pair indicator
75	Localised-orbital energy levels	Energy support for 26 node scalar reductions
76	Occupied-LMO atom counts	Count support for 13 occupied-LMO scalars
77	Occupied-LMO atom indices	Index support for 13 occupied-LMO scalars
78	Occupied-LMO coefficients	Coefficient support for 13 occupied-LMO scalars
79	Virtual-LMO atom counts	Count support for 13 virtual-LMO scalars
80	Virtual-LMO atom indices	Index support for 13 virtual-LMO scalars
81	Virtual-LMO coefficients	Coefficient support for 13 virtual-LMO scalars
82	Heat of formation	Two broadcast node invariant scalars
83	Point-charge dipole	Broadcast node Cartesian vector (T1)
84	Hybridisation dipole	Broadcast node Cartesian vector (T1)
Response derivation support		
85	Topology-aligned MOPAC bond orders	Bond-indexed Wiberg scalar; peptide-split support
86	Ring inventory	Structured ring table; derivation support
87	Ring atom membership and order	Ring-atom incidence table; derivation support
88	Aromatic-ring centres and normals	Centre + axial normal + radius; derivation support